

Phuc Sanh Nguyen

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Objective

Aspiring Full stack Developer and Software Engineering student at FPT University (GPA 3.6/4.0), focused on building clean, maintainable systems and clean architecture. 5 months of real-world experience: a 4-month OJT at FPT Software Academy building Java / Spring Boot services with a React front-end, and a 1-month on-site internship at FPT Japan (MDE Division), where I built a Japanese-Vietnamese dictionary desktop application on a self-written C11 language engine with Qt 6 / QML. Also experienced with C# / ASP.NET Core and Entity Framework Core through the EVCare platform. Seeking a Fresher/Junior position to contribute to production-grade code and write high-quality, testable software in a professional team.

Work experience

FPT Japan - MDE Division

May 2026 - Jun 2026

C / C++ & Qt Developer (On-site Internship - 1 month)

- Joined the MDE Division for a 1-month on-site program in Japan, working directly in an offshore team serving Japanese clients.
- Owned the development of an offline **Japanese-Vietnamese dictionary desktop application** built on a self-written C11 language engine with a Qt 6 / QML front-end — full technical detail in the Projects section below.
- Followed the customer's strict development process: detailed design documents, unit test cases with evidence, review checklists and defect tracking before release.
- Communicated with BrSE, QA and team leaders in Japanese (JLPT N4) and English; practiced the Ho-Ren-So reporting culture in daily work.
- Gained hands-on experience with the Japanese IT working style: documentation discipline, careful estimation and a quality-first mindset.

FPT Software Academy (FSA) - OJT Program

Dec 2025 - Apr 2026

Java / Full-stack Developer (On-the-Job Training - 4 months)

- Completed a 4-month OJT program building a full-stack internal management system with **Java 17, Spring Boot and React**.
- Designed and implemented RESTful APIs in a layered architecture (Controller - Service - Repository) using Spring Data JPA / Hibernate, DTO mapping with MapStruct and Bean Validation; tuned persistence by fixing N+1 queries with fetch joins and adding pagination / sorting to list endpoints.
- Implemented JWT authentication and role-based authorization with the Spring Security filter chain, plus centralized error handling via `@RestControllerAdvice` for consistent API responses.
- Built the React front-end with functional components and hooks, React Router, reusable form components, and Axios interceptors handling JWT attachment, token refresh and centralized error notifications.
- Wrote unit tests with JUnit 5 and Mockito and controller tests with MockMvc; fixed QA-reported defects and followed the team's coding conventions and SonarQube feedback.
- Worked in a Scrum team (sprint planning, estimation, daily stand-ups, sprint demo) and used Git / GitLab with feature branches, pull requests, code review and CI pipelines for build and test.

Projects

Japanese - Vietnamese Dictionary - Cross-platform Desktop Application

May 2026 - Jun 2026

FPT Japan - MDE Division

C / C++ & Qt Developer

- Built a complete offline Japanese-Vietnamese dictionary desktop app from scratch: a **C11 language engine** (*dict-core*) wrapped by a **C++17 / Qt 6 QML** application, with no runtime dependency beyond Qt and an embedded SQLite amalgamation.
- Architected the application in **MVVM with the Repository pattern**: QML views bind to a `QAbstractListModel` ViewModel that depends only on interfaces (`IDictionaryRepository`, `IUserRepository`, `ITextAnalyzer`, `ISpeechService`) injected at start-up — keeping business logic decoupled, mockable and testable.
- Implemented the Japanese language processing in pure C: UTF-8 code-point handling, romaji → hiragana conversion, automatic input-type detection (Japanese / romaji / Vietnamese), and a **de-inflection engine** that maps a conjugated verb back to its dictionary form (up to 16 candidate stems), so users can look words up exactly as they appear in real text.
- Wrote a **conjugation generator** that classifies verbs (godan / ichidan / suru / kuru) from their kana ending and part of speech and produces 14 grammatical forms (te, ta, nai, masu, potential, passive, causative, volitional, imperative, conditional...).
- Designed the data layer on **embedded SQLite compiled with FTS5 over 29,000+ dictionary entries**: prefix indexes on word / reading / romaji for instant lookup, an FTS5 virtual table for full-text meaning search in Vietnamese and English, and limit / offset pagination with total-count queries powering infinite scrolling.
- Separated dictionary data from user data with SQLite ATTACH: search history, favorites with multi-category tagging, personal notes and flashcards live in a private user database, so the dictionary can be rebuilt or updated without losing any user progress.
- Added a **flashcard review page** and Japanese **text-to-speech** through Qt TextToSpeech, using compile-time feature detection so the application still builds and runs gracefully when the speech module is unavailable.
- Developed a separate **CLI importer tool** (Qt Core) that streams 29,000+ rows from CSV into SQLite in a single transaction and then builds the indexes and the FTS index, cutting a full data rebuild down to seconds; the multi-target build (static core library, QML module, importer) is driven by CMake.

EVCare – Electric Vehicle Service Management Platform
Sep 2025 - Nov 2025

Fullstack Developer (Student Project)

- Developed a responsive and intuitive management dashboard using React and ensuring a smooth user experience across different devices.
- Implemented complex state management and utilized Axios to handle asynchronous API calls, including centralized error handling and JWT token persistence.
- Designed and implemented a 3-layer architecture (API, Application, Data) using Entity Framework Core with Repository and Unit of Work patterns.
- Built RESTful APIs for registration & login, JWT authorization filters by role (Admin, Staff, Customer, Technician), appointment booking, vehicle & service management, parts inventory, invoices and technician working sessions.
- Integrated online payments with VNPAY and PayOS; handled payment callback, order status updates and error cases.
- Used Hangfire to schedule background jobs and MailKit to send notification / reminder emails to customers.
- Collaborated with a 4-member team, defined API contracts with the React front-end, wrote unit and integration tests, and deployed via GitHub Actions (CI/CD) to Azure (backend) and Netlify (frontend).
GitHub: <https://github.com/HuongDS/EVCare>

Education

FPT University
2023 - 2027 (expected)

Bachelor of Software Engineering

- GPA: 3.6 / 4.0
- Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Web Development, Software Testing.
- Active in project-based learning across two backend stacks: Java / Spring Boot with JPA & PostgreSQL, and C# / ASP.NET Core with Entity Framework Core & SQL Server.
- Proficient in building responsive UIs with React and Next.js; experienced in state management, component lifecycle and integrating front-end applications with RESTful APIs.

Honors & Awards

2024	Second Prize (Co-Winner) – ICPC Vietnam Southern Provincial Programming Contest
2024	First Solve Award of F Talent Code
2025	Top 5 outstanding students in Spring 2025 at FPT University

Certifications

2025	JLPT N4
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Skill

Java 17 / Spring Boot / Spring Security / Spring Data JPA	C# / ASP.NET Core / .NET 6/8 / Entity Framework Core
React / JavaScript / Next.js / Axios	C / C++17 / Qt 6 / QML / CMake
SQL Server / PostgreSQL / MySQL / SQLite (FTS5)	RESTful API / JWT / Hangfire · Git / GitLab / CI-CD
Unit & Integration Test (JUnit 5, Mockito, xUnit, Moq)	Clean Architecture / Repository / MVVM · Agile / Scrum